

VIGIL-FLOOD: Village-level Integrated Geospatial & IoT Lead-Time Early Warning System

A Multi-Source Physics-Informed Predictive System for Coupled Flash Floods & Debris Flows in Mountain Catchments

Problem Statement ID:	SIH 26192	Organization:	Ministry of Home Affairs (MHA) / NDRF
Theme:	Disaster Management (Hilly Regions)	Pilot Area:	Beas River Basin, Mandi (Himachal Pradesh)
AI Model Architecture:	GPU XGBoost Ensemble + PyTorch Bi-LSTM	Lead Time:	10 to 30 Minutes Dynamic Action Window
Key Breakthrough:	Autonomous Sensor-Outage Fallback Engine	Status:	Production-Ready Full Prototype

EXECUTIVE SUMMARY: Traditional disaster forecasting fails in Himalayan hilly terrains because it relies on broad, district-wide meteorological forecasts that cannot resolve steep valley hydrodynamics or coupled landslide dams. **VIGIL-FLOOD** solves this by fusing 5 synchronous data streams: (1) Hourly rainfall accumulations, (2) Satellite & in-situ soil saturation %, (3) 30m DEM slope stability mechanics, (4) ISRO/GSI historical landslide inventories, and (5) Real-time IoT river telemetry. The system delivers hyper-local ward-level predictions, calculates a dynamic 10-to-30 minute actionable lead time, and auto-generates uphill evacuation routes to safe ridge shelters (>950m).

1. Official Problem Statement Analysis & Compliance

According to the Ministry of Home Affairs (MHA) and National Disaster Response Force (NDRF), hilly states in India (Himachal Pradesh, Uttarakhand, J&K;) face severe vulnerability to sudden flash floods and slope collapses. The official problem statement explicitly requires a multi-source predictive system generating ward-level forecasts with actionable lead time.

NDRF Requirement	Underlying Hazard Physics	VIGIL-FLOOD Implementation
Hilly states vulnerable to landslides & flash floods	Himalayan flash floods are coupled debris flows (slurry density ~2.0 g/cm ³ , speed 20–40 km/h).	Coupled Dual-ML Engine: Evaluates both river catchment overtopping and steep slope shear instability simultaneously.
Integrates rainfall, soil moisture & slope stability	Soil saturation exceeding 85% eliminates internal friction, triggering slope shear failure ($F_s < 1.0$).	Physics-Informed Features: Computes Infinite Slope Factor of Safety (F_s) and Rational Runoff ($Q = C \cdot I \cdot A$).
Historical landslide inventories & IoT inputs	Ancient geological slip planes reactivate under prolonged multi-hour monsoon soaking.	ISRO/GSI Weighting + IoT: Combines Bhuvan landslide scar densities with live ultrasonic river gauges.
Hyper-local village/ward forecast & lead time	Valley basins suffer total inundation while high ridges remain stable above the waterline.	Dynamic Lead-Time Engine: Computes exact evacuation windows (10–30 min) and routes to shelters >950m.

2. The 4-Phase Himalayan Flash Flood Debris Cycle

In steep mountain valleys, flash floods are not pure water; they are hyper-dense, boulder-carrying debris flows. Our system models the complete 4-phase physical lifecycle of a mountain disaster:

Phase	Trigger Mechanism	Geomorphic Impact & Equations
Phase 1: Saturation Trigger	Cloudburst precipitation (>50 mm/h) saturates soil moisture past 85–90%.	Pore-water pressure spikes; soil shear strength $\tau = c' + (\sigma - u)\tan\phi'$ drops toward zero.
Phase 2: Slope Shear Failure	Factor of Safety drops below 1.0 on steep hillsides (>30°).	Thousands of tons of mud, boulders, and trees collapse into mountain tributary streams (nallahs).
Phase 3: Valley Gorge Damming	Landslide debris chokes narrow V-shaped river bottlenecks.	A temporary geological dam forms, impounding millions of liters of water within 10–20 minutes.
Phase 4: Hydraulic Breach & Surge	Debris dam fails catastrophically under hydraulic head pressure.	A high-velocity slurry ($E_k = \frac{1}{2}mv^2$, $\rho \approx 2.0 \text{ g/cm}^3$, 20–40 km/h) destroys downstream low-lying riverside wards.

3. Multi-Source Data Ingestion & Satellite Resources

VIGIL-FLOOD eliminates data silos by ingesting and harmonizing 5 data sources across free, verified global and Indian repositories:

- **Rainfall Accumulation:** Open-Meteo & OpenWeather Live APIs — Provides hourly precipitation (mm/h), 3h/6h/24h rolling totals, and 3-hour forecast rates (10,000 free calls/day, zero registration).
- **Soil Moisture Saturation:** NASA SMAP & Copernicus ERA5-Land — Topsoil volumetric soil water (0–7cm) converted to % saturation (0–100%) to calculate pore-water pressure.
- **Topography & Elevation (DEM):** Copernicus GLO-30 & NASA SRTM 30m — High-resolution 30-meter elevation raster providing terrain slope angles (10° to 55°) and flow accumulation channels.
- **Landslide Inventories:** ISRO Bhuvan Landslide Atlas of India & GSI NLSM — Historical slip scars across Mandi and Kullu districts to weight baseline susceptibility.
- **Real-Time IoT Telemetry:** Ultrasonic River Level Gauges (JSN-SR04T) & MEMS Tiltmeters (MPU6050) — Streamed via WebSockets/MQTT every 3 seconds for microsecond response.

4. Production AI Architecture & GPU Benchmarks

Trained across **280,512 real hourly records (2021–2024)** spanning the Himalayas (Mandi, Beas Basin), Japanese Alps (Nagano), and Austrian Alps (Innsbruck) on an NVIDIA CUDA GPU:

Model Name	Target Hazard	Feature Schema	Training Time	R² Accuracy	RMSE
FULL_INUNDATION_MODEL	Flash Flood Inundation	13 Features (All Sensors)	1.04 s (GPU)	99.98%	0.0008
FULL_SLOPE_MODEL	Landslide / Debris Flow	13 Features (All Sensors)	1.05 s (GPU)	99.99%	0.0014
FALLBACK_FLOOD_MODEL	Sensor Outage Mode	11 Features (No Water Sensor)	0.86 s (GPU)	99.98%	0.0009
FALLBACK_SLOPE_MODEL	Sensor Outage Mode	11 Features (No Water Sensor)	0.92 s (GPU)	99.99%	0.0013
PYTORCH_BiLSTM_MODEL	Temporal Storm Trajectory	6-Hour Sequence Windows	72.5 s (10 Epochs)	90.30%	0.0007

5. 3D Mountain Terrain Digital Twin & Simulation

VIGIL-FLOOD features a 100% free open-source 3D WebGL Digital Twin (MapLibre GL JS + AWS Terrarium 3D DEM with 1.6x vertical relief):

- **Real 3D Mountain Mesh:** Renders deep V-shaped Himalayan gorges with true elevation heights, 65° camera tilting, and 360° rotation.
- **Real Google Hybrid Imagery:** Drapes high-resolution Google Satellite tiles over the 3D relief, displaying real village names, streets, and NH-3 Highway.
- **Dynamic 3D Flood Inundation Extrusion:** When rainfall or river surge sliders are adjusted, the 3D water layer physically swells, widens, and rises up the canyon walls in real-time right before your eyes!
- **Click-to-Get Spot Elevation:** Clicking any 3D mountain peak or slope queries Copernicus 30m DEM in real time (<50ms), returning exact altitude and terrain vulnerability classifications.

6. Autonomous Sensor Fault-Tolerance (The Killer USP)

During extreme mountain cloudbursts, river-level IoT sensors are frequently broken or washed away by rolling boulders. In traditional early warning systems, missing sensor data causes the entire software pipeline to crash.

The VIGIL-FLOOD Solution: Our backend Sensor Health Manager continuously monitors incoming telemetry heartbeats. The millisecond an outage is detected on a river gauge, the system autonomously dispatches the **11-Feature Fallback Model**. It reconstructs hydrological risk from rainfall intensity, soil saturation %, and 30m slope runoff, preserving the critical evacuation alert with **74.0% to 99.9% confidence and zero system downtime!**

7. Last-Mile Action & Citizen Evacuation Routing

A prediction is useless without actionable evacuation guidance. VIGIL-FLOOD bridges the last-mile gap:

- **Dynamic Lead-Time Engine:** Calculates actionable escape windows in minutes (10–30 min) based on distance to riverbed, runoff acceleration (Q), and rate of water rise (dH/dt).
- **High-Ridge Safe Shelters:** Automatically assigns citizens to pre-surveyed high-altitude concrete buildings (e.g. Govt Senior Secondary School at 990m, safely +110m above the 880m flood basin).
- **True Uphill Safe Routing:** Calculates escape paths strictly climbing uphill along mountain ridge roads, preventing panicked villagers from walking into low-lying riverside flood traps.
- **Common Alerting Protocol (CAP) SMS:** Auto-generates multi-lingual geo-targeted emergency SMS alerts for instant telecom broadcast to village pradhans and residents.

8. 10-Minute Presentation Script & Jury Defense

Slide-by-Slide 10-Minute Presentation Pitch Script:

- Slide 1 (1.5 min) — The Himalayan Crisis:** Explain why mountain flash floods are 20–40 km/h lethal debris flows, not flat slow-rising city water. Introduce VIGIL-FLOOD as the hyper-local village early warning system.
- Slide 2 (2.0 min) — 5-Source Multi-Sensor Architecture:** Present the synchronized data pipeline fusing rainfall, satellite soil moisture, 30m DEM slope physics, ISRO landslide history, and real-time IoT.
- Slide 3 (3.0 min) — Live Command Center & 3D Digital Twin:** Demonstrate Pandoh in 3D: show the Beas River canyon, click for 30m DEM elevation (880m vs 990m), drag the cloudburst slider, and watch the 3D water physically flood the valley in real time!
- Slide 4 (2.0 min) — Sensor Outage Fault-Tolerance (The USP):** Click 'Disconnect Water Sensor' live on screen. Show judges how the system auto-switches to the Fallback Model and preserves the evacuation alert with 74%

confidence!

Slide 5 (1.5 min) — Citizen View & Last-Mile Evacuation: Switch to Citizen View: show high-ridge shelter routing, dynamic 10–30 min lead times, and automated CAP emergency SMS broadcasts.

Top 5 Tough Jury Defense Answers:

Q1: Why did you include landslides in a flash flood system?

Answer: In the Himalayas, flash floods are coupled debris flows. Cloudburst rain saturates soil past 85%, causing landslides that choke river gorges. When the dam breaches, it releases a high-velocity slurry ($E_k = \frac{1}{2}mv^2$, $\rho \approx 2.0 \text{ g/cm}^3$). That is why NDRF explicitly mandates slope stability models!

Q2: How do you calculate Actionable Lead Time?

Answer: Based on four dynamic variables: (1) distance from village settlement to riverbed, (2) DEM slope runoff acceleration, (3) rate of river rise (dH/dt), and (4) soil saturation velocity.

Q3: What if internet connectivity fails in mountain valleys?

Answer: VIGIL-FLOOD is designed for edge-computing. The lightweight models can run locally on an edge IoT gateway (ESP32/Raspberry Pi) at the village Panchayat Bhawan, triggering local siren alarms and mesh radio alerts.

Q4: How is this different from standard IMD forecasts?

Answer: IMD provides broad district-level 24-hour rainfall forecasts. VIGIL-FLOOD provides hyper-local ward-level predictions (50–100m resolution), identifies exact submerged houses, and computes minute-by-minute lead times.

Q5: How did you train and validate your models?

Answer: Trained dual XGBoost and PyTorch Bi-LSTM models across 280,512 real hourly records (2021–2024) spanning the Himalayas, Japanese Alps, and Austrian Alps on an NVIDIA CUDA GPU, validated with SHAP feature attribution.

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